Answer key: Lesson 4, Cubes and Views

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Look at the stack of cubes.

Answer: 5 cubes

Touch each cube once: 5 cubes. No cube is hidden in this picture.

Watch for: Counting a cube twice.; Counting the faces you can see instead of the cubes.

S2. The stack has 4 cubes in all.

Answer: 3; 1

You can see 3 cubes in the picture. The stack has 4 cubes, so 1 cube is hidden. 3 and 1 make 4.

Watch for: Writing 4 for the cubes you can see.; Writing 0 because nothing looks hidden.

S3. Which top view matches this stack?

Answer: B

The back row of the stack has 1 cube and 1 cube. The front row has 2 cubes and 1 cube. So the top row is 1, 1 and the bottom row is 2, 1. That is picture B.

Watch for: Choosing the view with the rows the wrong way up (A).; Putting the tall stack in the wrong place (C).

S4. Look at the stack. Write how many cubes are in each stack on the top view.

Answer: 1 and 3 in the top row; 1 and 1 in the bottom row

The back row has a stack of 1 and a stack of 3. The front row has a stack of 1 and a stack of 1. The top view is 1, 3 on the top row and 1, 1 on the bottom row.

Watch for: Writing the rows the wrong way up.; Writing the number of faces you can see.

S5. A paper square is folded so that the top edge meets the bottom edge.

Answer: A line across the middle, from the left edge to the right edge

When the top edge meets the bottom edge, the paper bends halfway down. The fold line goes across the middle of the square, from the left edge to the right edge.

Watch for: Drawing the line down the middle.; Drawing a line across a corner.

S6. The top view shows how many cubes are in each stack.

Answer: 7 cubes

2 + 3 + 1 + 1 = 7. There are 7 cubes in the stack.

Watch for: Adding only two of the numbers.; Counting the squares (4) instead of the cubes.

Think harder (page 3)

T1. The stack has 6 cubes in all.

Answer: 2; 1

The top view has 2, 1 and 1. 2 + 1 + 1 = 4. The stack has 6 cubes, so the missing number is 2. The picture shows 5 cubes, so 1 cube is hidden.

Watch for: Writing 6 for the missing number.; Saying no cube is hidden.

T2. A paper is folded in half. The fold is on the left. One hole is punched.

Answer: 2; second

The fold is in the middle of the opened paper. One punch goes through 2 layers, so there are 2 holes, one each side of the fold and the same distance from it. That is the second picture. The first has holes on the same side. The third has only 1 hole.

Watch for: Writing 1 hole.; Choosing two holes on the same side of the fold.

T3. A paper was folded along the dashed line and one hole was punched. The picture shows the opened paper. One hole is ringed.

Answer: 1 step on the other side of the dashed line

The ringed hole is 1 step to the left of the dashed line. The other hole is 1 step to the right of it, on the same row. Draw a hole on that dot.

Watch for: Drawing the hole on the same side.; Counting dots instead of steps.

Go further (page 4)

G1. A paper square is folded in half, then in half again. One hole is punched through all the layers.

Answer: 4; third

After the first fold the paper has 2 layers. After the second fold it has 4 layers. One punch goes through 4 layers, so there are 4 holes, one in each part of the opened paper. That is the third picture.

Watch for: Writing 2 holes.; Choosing 4 holes in a row.

G2. A stack has a 2 by 2 base. The back stacks have 3 and 1 cubes. The front left stack has 2. The stack has 8 cubes in all.

Answer: 2; 1

3 + 1 + 2 = 6. 8 take away 6 is 2, so the front right stack has 2 cubes. The picture shows 7 cubes, so 1 cube is hidden behind the front right tower.

Watch for: Writing 8 for the front right stack.; Saying no cube is hidden.

G3. Build a stack with 7 cubes on a 2 by 2 base. No stack is taller than 3 cubes. Many answers are right.

Answer: Many answers are right: four numbers from 1 to 3 that add to 7, for example 3, 2, 1, 1

For example 3, 2, 1 and 1: 3 + 2 + 1 + 1 = 7. Another answer is 2, 2, 2 and 1. Every stack has at least 1 cube.

Watch for: Four numbers that do not add to 7.; A stack taller than 3.

Investigation (page 5): Fold, mark and open

There will be 2 dots. The two dots are the same distance from the fold line, one on each side. The paper had two layers, so one dot made two.

Look for: A child who says the dots are the same distance from the fold, or are opposite each other across the fold line. Credit any clear reason. A child who folds again and finds 4 dots has gone further.

Exit ticket (page 5)

E1. Look at the cubes.

Answer: 4 cubes

Touch each cube once: 4 cubes. No cube is hidden in this picture.

Watch for: Counting a cube twice.; Counting faces instead of cubes.

E2. The top view shows how many cubes are in each stack.

Answer: 6 cubes

2 + 1 + 2 + 1 = 6 cubes.

Watch for: Counting the squares (4).; Adding only the top row.

E3. A square is folded so the left edge meets the right edge.

Answer: down the middle

When the left edge meets the right edge, the paper bends halfway across. The fold line goes down the middle of the square, from the top edge to the bottom edge.

Watch for: Choosing across the middle (that is where the top and bottom edges meet).; Thinking the fold is at one side.

Sneak-peek puzzles (not in the workbook)

1. 4 cubes.
2. The 5th cube is hidden behind the front tower.